2: Discipline

Science, Sociology, and Their Core Concepts

Introduction

Here is one (very negative) view about the current state of Sociology:

> According to Ray Rodrigues, chancellor of the state university system, Florida lawmakers began eyeing general education as a target of reform after complaints from parents. Their children would come home for Thanksgiving break during their first semester in college and would "be communicating cultural Marxism"...Under SB 266 [a Florida state law intended to eliminate "woke" ideology], courses that count toward general-education credit should "promote and preserve the constitutional republic" through "traditional, historically accurate" coursework. The law also says that "core courses" may not "distort significant historical events," "include a curriculum that teaches identity politics," or be based on theories that systemic racism, sexism, oppression, and privilege are inherent in American institutions.[1]

There is much to unpack in this statement. For now, we will leave aside the meaning of "cultural Marxism."[2] We will also only briefly note that colleges explicitly advertise themselves as places to "push boundaries, challenge assumptions and redefine what's possible,"[3] so one would *expect* young people to *want* to discuss new and interesting ideas with their family. We will also only glancingly note that "promot[ing] and preserv[ing] the

[1] "Is Sociology Being Saved — or Debased — in Florida?" *Chronicle of Higher Education*, March 16, 2026.

[2] I seriously doubt that Rodrigues and his allies are reading Gramsci, Lukacs, or Benjamin, and instead, are simply using "cultural Marxism" as a catch-all for ideologies they don't like. See A.J.A. Woods, *The Cultural Marxism Conspiracy*, Verso, 2026.

[3] Stony Brook's Webpage, accessed June 27th, 2026.

constitutional republic" is *itself* an ideological goal, and that pursuing it explicitly could and would bias our understanding of society just like an explicitly "woke" one would.

Instead, I want to take aim at what seems to be a foundational assumption of those criticizing sociology: that there is a way to pursue a completely unbiased, objective, neutral study of society. The authors of the Flordia state law imply, by claiming to cleanse University curriculum of "distortions," that there *is* an objective, true, and indisputable nature of society and history that is like the things we know about the physical history of the universe, biological dynamics, or the engineering of the material environment.

As you will see below, at a very basic level sociology (and the rest of the social sciences) do not work this way. This is because they take as their central object *people in society*, and since people *react* to their understandings about the world and the meaning they take from it, they need to be studied in a different way than rocks, planets, and pathogens.[4] However, this does *not* mean that the social sciences and particularly sociology are therefore simply a realm where we assert opinions and call them facts; instead, it means that we have a wide inventory of careful, systematic ways of understanding the world, as well as many key concepts and theoretical traditions, and our job here in part is to introduce them.

[4] As noted in the previous unit, this is Du Bois' notion of **reflexivity**.

What Is A Science?

To answer the question "what kind of science is sociology?", we need to answer a prior question: just what is a **science**? And to do *that* we have to take yet another step backwards, and, in turn, ask two basic questions about the structure of the world:

1. What kinds of things make up the world?
2. How do we know what those things are?

These are both ultimately philosophical questions. The first is a question of **ontology**—sometimes also called "metaphysics"—and represents our best understanding of not only what the smallest, most basic parts of the world are, but also how those parts combine into all of the things that exist. The second question, meanwhile, is one of **epistemology**; *if* the world is populated with certain things, how can we know what they are?

Since the emergence of modern science about 500 years ago (depending on how you're counting; see Shapin (1994)), a strong consensus has grown around two epistemological commitments. The first is that all explanations and descriptions of things in the world—*what* is there, *how* those things interrelate, and *how* we know—must be **natural**. Bluntly, this means that scientists aim to explain things in the world as the result of the workings of other observable things in the world, and not as the result of the will of one or another deity.

Ontology is the study of the basic building blocks of the natural world. **Epistemology** is the study of ways of knowing about and verifying the accuracy of our knowledge about the world.

The second point of consensus among all forms of science is a (more-or-less) **skeptical attitude** towards existing explanations about the world. This means that, rather than accepting a description or explanation about how things work in the world just because they come from someone we respect or fear, we instead try to objectively verify that a given process *actually* works that way. Crucially, this does not mean that *anyone* is qualified to question scientific findings (since they often depend on years of experience and training to understand extremely complex processes), but rather that communities of scientists should work as hard as they can to subject all explanations to the most rigorous scrutiny they can imagine (Merton 1979).

Naturalistic explanation is a key aspect of scientific epistemology. It dictates that all phenomena subject to scientific description and explanation should relate only to processes and things in the world, not to supernatural forces.

As mentioned, commitments to skepticism and naturalism are key parts of what defines a science. But there are also important tensions and divergences within science. Three of the most central to sociology are worth addressing in some depth.

In scientific epistemology, the **skeptical attitude** is a commitment held by scientists to subject potential descriptions and explanations about the world to the most rigorous examination possible to verify them.

The Problem of Realism vs. Empiricism

First and foremost, a central epistemological tension in science concerns the observability of phenomena under study. Most basically, things can be observed directly with our senses—we can see, hear, feel, touch, and taste what we want to know about. At the level of our raw, physical experience of the natural world, this gives us a lot of information necessary to survive. For example, I know my coffee cup won't fall through the table right now. Why? Because, before I put it down, I can verify that the table has a physical property of being "hard" by touching it, and therefore be (pretty) sure that this will persist as long as the cup is there. Philosophers of science call this epistemological approach **empiricism**.

But there is *large* class of phenomena—containing most of what sociologists care about—that are not readily accessible to an individual's direct observation in this way. For instance, think of THE MARKET.[5] Yes, you *can* go down to Smith Haven Mall, or even just to the food trucks collected outside the Student Activities Center, and see *parts* of THE MARKET. But what prices you will pay in either of those places, and, indeed, what will even be available to you, are actually driven by a world-spanning social structure.[6] So those two places are *a part of* THE MARKET, but only parts, and there is literally no way to observe the whole thing; instead, scientists can only observe the *traces* and *parts* of it that *are* available to observation, and then work backwards, asking themselves "how would THE MARKET have to work for us to see prices, the availability of goods, and so on work in this way"?

[5] I'm using smallcaps to denote THE MARKET, in the sense of the real social structure, and "markets" to denote concrete venues for buying and selling.

[6] For the food trucks, this is the "global commodities market" for foodstuffs and other materials, which indirectly set the prices the vendors pay to their wholesalers, and which thus determines whether they can profitably offer you a tasty, tasty burrito. If wholesale prices become too high, they must either give you less of the best parts of the burrito (obviously the pulled pork) for the same price—"shrinkflation"—or substitute something less expensive.

In this case, THE MARKET *is* real and a part of the world, since it has effects we can observe and measure, and philosophers of science therefore call the process of reasoning backwards from those *observable effects* to the processes that cause them **realism**.

Realism is the scientific approach of reasoning backwards from observable traces of a phenomenon to assess its structure and general effects in the world.

Now, if you're thinking that there is a *large* gulf between these two positions, you're right! A key way that scientists bridge the "directly observable" with "real but unobservable processes" is through **instrumentation**, or reliable machines and procedures that allow them to collect measurements that are only indirectly accessible.

Scientists use **instrumentation** to gather more precise measurements and observations about the operation of phenomena that are not ordinarily available to human senses.

Continuing with the example of THE MARKET above, let's say that economists and economic sociologists want to know whether prices are generally going up or down for consumers. In the United States, we have a government agency, the Bureau of Labor Statistics that conducts a survey called the "Consumer Price Index", which

> …is a measure of the average change over time in the prices paid by consumers for a representative basket of consumer goods and services. The CPI measures inflation as experienced by consumers in their day-to-day living expenses.
>
> The CPI represents all goods and services purchased for consumption by the reference population. BLS has classified all expenditure items into more than 200 categories, arranged into eight major groups (food and beverages, housing, apparel, transportation, medical care, recreation, education and communication, and other goods and services). Included within these major groups are various government-charged user fees, such as water and sewerage charges, auto registration fees, and vehicle tolls.

The BLS works extremely hard to construct a sample that *accurately* measures the "real" CPI, but the point for our example here is that they are seeking to measure a *general property of the consumer goods market in the US*, and not any individual, particular price. In sum, you could not obtain this description of the behavior of THE MARKET in the US without the instrument of this survey, or something like it.[7]

[7] Incidentally, most "hard" sciences work like this too. Take the Nancy Grace Roman Space Telescope, soon to be launched. It works by scanning the infrared spectrum of light, which you can't see with your eyes, but which is well-enough verified and understood that it has widespread applications today.

The Problem of Emergence vs. Reduction

The tension between realism and empiricism is heavily epistemological, but another tension in science involves ontology, and can be understood through instrumentation. Most scientific instruments are built (and validated) using assumptions about what kinds of objects they are searching for. To continue with our example above of THE MARKET, if we want to know its properties, it is probably not a good idea to interview a single consumer and ask about their feelings regarding prices.[8] Yet at the same time, the behavior and properties of THE MARKET are driven by the actions of innumerable people acting (according to classical economics)[9] simply trying to maximize their own local interests.

[8] Scientists do, however, survey the *collective* sentiments of populations about the market. The most famous is done by the University of Michigan.

[9] The classical statement of this perspective of the "invisible hand," which vastly oversimplifies the perspective of contemporary economics, is from Smith (1976).

But which of these two perspectives—collective or individual—shows the "true" nature of THE MARKET? This is not a philosophy class, so thankfully, we don't have to solve this ontological question.[10] Instead, it reveals that different scientific disciplines are at least partly defined by which ontological level they assume is fundamental to their studies.

[10] And indeed, how could we? My own approach is pragmatic—different ontological perspectives are useful to do different things, and the question is what each allows and forbids, not which one is correct for all time. See Hacking (1983).

At one extreme, some disciplines try to **analytically reduce** their objects of study to those that they purport to be the most fundamental in the universe. Thus, the last time I checked, physics recognizes seventeen fundamental particles, and literally everything everywhere is composed of them.[11]

[11] Fascinatingly, while this is the most reductive ontology that has achieved widespread scientific consensus, it has deficiencies, and there are rumblings of even further reductions.

Analytic reduction is the assertion that complex phenomena are actually collections of constituent parts, and therefore best studied in terms of those smaller components.

Analytic reduction can be extremely powerful and it has revealed some of the most fundamental mechanics of the world around us that we know of. But also think of the phenomena in the world that analytic reduction makes it *harder* to study. Let's go back to THE MARKET, for instance. *If* you ~~were insane and~~ wanted to study THE MARKET in terms of quantum physics, you would *at least* have to make the following steps:

1. Explain how quantum particles become atoms.
2. Explain how atoms become molecules.
3. Explain how inorganic molecules become organisms.
4. Explain how organisms' neural functions become sufficiently complex to attain conscious and intentionality.
5. Explain how consciousnesses form collective abstract representations.
6. Explain how those representations evolve and grow more complex to become the modern form of THE MARKET.

To my knowledge,[12] only one of these steps (step 2) is well-understood, and if you tried to move across all 6, you would easily exhaust your own life trying to just explain *one* complex phenomenon.

[12] Those of you who know more than me can tell me whether the other steps have been decisively accounted for.

By consequence of the difficulty of working all the way back up from purportedly fundamental elements to complex phenomena, many sciences focus on the properties that **emerge** at a given level of complexity. In other words, for THE MARKET, you don't need to know the quantum-physical properties of the gold being sold in (one part of) it, because they are irrelevant to what you are trying to explain—say, the availability of gold at a certain price and at a certain time. Likewise, a clinician who suspects one of their patients is subject to intimate partner violence is unlikely to be concerned by the organic chemistry of their patients' injuries; instead, they will likely focus on the psychological and social processes at work.

Throughout this discussion, you'll notice that I have not once asserted that any particular ontological level is "right," nor that reduction or emergence is always the best approach. But I also want to emphasize that the boundaries among areas of research are not at all immutable. Indeed, some of the most interesting areas of science today are at the boundaries between reduction and emergence, and protective "border skirmishes" between disciplines tend to be fought over them.[13] For instance, a great deal of work is being done trying to understand the intersection between the biochemical processes in the brain and our sense of ourselves as conscious beings. And an emergent field of study is asking whether LLMs—trained on our collective discursive output—can be studied directly as repositories of social dynamics not evident in any given piece of discourse. Instead, the ontological levels assumed by given disciplines act like centers of gravity around which most of the work considered to be "in" that discipline orbits.

Emergence is the unfolding of new properties and higher levels of ontological complexity, which can be apprehended and studied without reference to subsidiary ontological levels.

[13] For example, *sociological* social psychology tends to criticize *psychological* social psychology as having an impoverished account of the emergent properties of social structure and culture, whereas in the opposite direction, psychologists see sociologists as having woefully imprecise concepts of "the self," "personality," "motivation," and so on—which is to say, they have insufficiently analytically reduced their concepts to be useful.

The Problem of Subjectivity, Intersubjectivity, and Objectivity

A final tension in science worth discussing involves the source of observations about the world—people themselves. People are both ontological and epistemological problems for science.

Ontologically, they represent something that is maddeningly hard to observe, which is interior self-conscious states and intentional motivations about the future. This is usually summarized as their **subjective**. Worse still, people are also capable of *coordination*, which is to say they can exchange meaning[14] with one another to align themselves into regular patterns, or even to *change* the state of the world through collective action to bring about something they imagine to be possible. Scientists usually call this kind of exchange and coordination **intersubjective**. And if even *that* were not complex enough, there are also sets

[14] The subject of unit 4!

of social and material forces working on whole populations, the planet itself, and physical processes that lie beyond the reach of even the most elaborate coordinated action we have ever observed. It is these phenomena we usually call **objective.**

Each of these three ontological levels also carries epistemological challenges for scientists. Scientists usually try to avoid purely **subjectivist** interpretations, which is to say, those than depend on an individual's perspective—or those belonging to one social category and not another—and aim to gather knowledge that transcends such narrow perspectives (or, at least, can be translated across perspectives).

A **subjective** ontology emphasizes people's interior states of mind, beliefs, and intentions about the world around them. An **intersubjective** ontology concentrates on the shared beliefs, goals, and intentions held by groups of people coordinating their action. And an **objective** ontology emphasizes how non-human material and large-scale social forces affect society and individual people.

A **subjectivist** epistemology focuses on people's internal mental states and how they influence people's actions and beliefs about the world.

However, it is also the case that one's subjectivity, when used reflexively, can be a key mode of gathering knowledge about the world. To take a trivial example, it is well-known that some people experience white coat hypertension where, because they are nervous to have their blood pressure taken in a hospital setting, their blood pressure measures higher than it would otherwise. Depending on what you want to know, the patient's subjectivity—that is, their sense of themselves and the situation given their social position in a given setting—is either a source of scientific bias (about their "true" blood pressure) or a phenomenon worth directly studying itself.

There are also **intersubjectivist** epistemological tensions in science. On the one hand, as noted above, the apparatus of modern scientific instrumentation means that to be able to meaningfully contribute to science often takes *years* of specialized training, leaving only a tiny proportion of the overall population qualified to skeptically examine scientific claims.[15] At the same time, to the extent that intersubjective scientific communities become locked into a single collective perspective, the very boundaries of expertise can also become blinders, and crucial phenomena in the world can be left under- or unstudied.

[15] I have no scientifically-sound idea whether the better avenue for pursuing controlled nuclear fusion power is a Tokamak Reactor or a Stellarator. Do you?

Partly in response to the threats of subjectivist and intersubjectivist bias, some scientists harken for an **objectivist** epistemology. Here, the best knowledge is that cleansed of the corruptions of individual moral or political commitments biasing the conduct of research, and from being "steered" towards one or another topic or conclusion by a disciplinary community. At a minimum, these scientists believe, this should mean that scientists commit

An **intersubjectivist** epistemology emphasizes how scientific perspectives, topics of inquiry, and interpretations are shaped by disciplinary experts in communication with one another.

to reporting their findings even if they show the opposite of what they hoped, or that scientific communities should protect or even reward conducting work on unpopular or controversial topics. At maximum, however, objectivist epistemology seeks a "view from nowhere" (Nagel 1989), completely detached and independent from any social influence.

One tension in this objectivist epistemology should be obvious: there is *no* such thing as a position outside of society from which to study it, and plenty of social phenomena are constituted depending on concrete social positions only accessible to certain scientists. To this objection, defenders typically respond that instead of a purely objective ontological view, we should instead *aspire* to being as objective as we can, and that a crucial step in that direction is to deliberately admit and reward those with views outside of the scientific mainstream into the intersubjective conversation.

An **objectivist** epistemology seeks to eliminate subjective and intersubjective bias from inquiry, and ultimately achieve a totally independent view of reality.

Key Concepts and Theories in Contemporary Sociology

So now we have dined on some philosophy—hopefully without choking!—and are ready for another course, this time at the sociological buffet. Here, my goal is to serve you a taste of what we will savor through the rest of the course: concepts that we will see over and over. I also hope you'll see how their flavors are all the richer for the philosophical background about science that we have just covered![16]

[16] Okay, okay, I'll stop beating the "food" metaphor to death!

Key Concepts

Social Construction

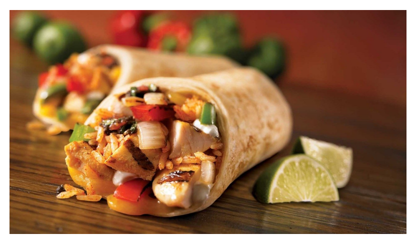
One of the simplest and most direct, but also most notorious, concepts in sociology is **social construction**. If you have ever played any kind of game, engaged in make-believe with a young child, read fiction, or traveled in an unfamiliar culture, you have encountered social constructions and been aware of doing so. But *also*, if you ever used a bank, been in love, celebrated a birthday, or spoken to another person, you've also encountered a social construction, but perhaps with less awareness. Simply put, the concept means that a phenomenon in the world exists because of shared belief in its existence, and people behaving as though it exists.

Crucially, this does not mean that individual people can simply make up whatever they want or that nothing in the world is "real."[17] Instead, it means that most of what we treat as real and self-evident for the purposes of getting through our everyday lives is an unproblematic *composite* of objective, material matter and the meanings we attach to it. We can (usually) treat the currency we hold in our hands as "good" to exchange for something else, and don't think twice about the fact that (US) currency is actually fiat money, and ultimately depends on global trust in the soundness of the US economy and treasury. And if we are using a credit card or our phones, we (usually) don't even think about the fact that the "dollars" we are spending are in fact simply electronic entries on various ledgers being changed, or that currency in circulation is a very different thing (much smaller) than the total money supply. We just buy that tasty, tasty burrito.[18]

Social construction is when a phenomenon is constituted by shared belief in its existence.

[17] For a lovely exposition of a sophisticated constructionist view, see Hacking (2004).

[18]



Micro and Macro

Sociology as a whole is usually concerned with intersubjective phenomena, but that intersubjectivity can take place at (at least) two very different scales. One is at the level of face-to-face interaction, and, indeed, a great deal of communication takes place in these venues: we shake hands, eat together, smile or frown at one another, laugh, cry, yell, fight, love, and so on. When we are talking like this about our interactions with the immediately-present world and interactions within a given social setting, sociologists describe this as the **micro** level of analysis.

But as I've alluded to already, many phenomena sociologists care about exist at a higher level than just face-to-face interactions. Think about "Stony Brook University," for instance. It's not *just* the buildings, faculty, students, and staff that make it

Micro-level analysis focuses on face-to-face interactions with other people and interactions with the immediately present world.

up; instead, it is an "organization" that enforces durable patterns, and allows us to operate at a surprisingly-efficient level of coordination. Put differently, even if *one* section of SOC 105 was cancelled, SBU would continue to be a university, but at the same time, it could not be a university without having *some kind of* classes. Taken together, a huge amount of the apparatus of modern life works like this, and collectively, sociologists call this level of analysis **macro**.

One key area of sociological research today is how these two levels intersect (Martin 2009), and whether or not they represent different ontological levels of social life (Hedstrom 2005).

Macro-level analysis focuses on scales of phenomena beyond the level of face-to-face, direct interaction with the world.

Agency and Structure

At the level of our personal lives, we all have choices. And when we organize collectively, those groups, too, can choose one course of action or another. In turn, these choices can make a difference in our individual lives, or even on the fates of whole organizations or societies. Sociologists call this process of making a difference in the world **agency** and the entities doing so **agents**.

Meanwhile, much of the social world also appears to us as constraint—it seems that our lots are set by factors outside of our control. For instance, it would nonsense in contemporary society for you to judge someone by saying "they should have picked their parents better!" This is because, of course, who we are born to (and, sociologically, the social class, national origin, race, and so on) is not chosen by us, but rather something we encounter as a given, and then must make our choices within that framework. We call these kinds of factors in the world **structures** and usually thing of them as determining individual or collective social outcomes.

Agency is the capacity of some entity to make a difference in the world.

Structures are patterned social phenomena that dtermien individual or collective outcomes.

Social Order and Social Change

Finally, when considering social structures over time, sociologists remark upon two main dynamics. First, and perhaps surprisingly, **social order** seems generally resilient, or at least resistant, to total destruction. No matter whether it is a war, natural disaster, or other calamity, *some* form of patterned collective activity based on a shared understanding of the situation seems to emerge.

That being said, specific social structures sometimes change, and sometimes they do so quite rapidly. For instance, within *your* lifetimes, the cheap, easily available technology has emerged and diffused which allows nearly instantaneous global communication.[19] This has massively changed how we interact with one another, above all, pushing that interaction into new digital spaces.

[19] This is, of course, the smart phone.

This rapid technological change is just one more feature of the long-term rise to dominance of capitalism as the key mode of organizing our economic lives. After all, the rise and diffusion of smartphones, and the spread of the internet as more than simply an investment by the military, required the ability to produce, market and sell phones and internet services as **commodities** at a profit. (The spread of capitalism has had numerous other effects which we will trace throughout the course.)

A second, and no less important, world-historical change is loosely related to the emergence of capitalism, and involves the use of calculation to think about an increasing number of aspects of our lives. This process, **rationalization**, entails *both* the subjection of action to means-ends efficiency logic (in which case it is called **formal rationalization**) *and* the articulation of explicit purposes for action in more and more domains of life (where it is called **substantive rationalization**).

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